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Modified gradient neural networks for solving the time-varying Sylvester equation with adaptive coefficients and elimination of matrix inversion

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ABSTRACT

In scientific and engineering fields, the solutions to many problems can be transformed into finding the solutions to Sylvester equation, for which various computational methods (e.g., recurrent neural network, RNN) have been presented and investigated. RNN models are frequently used to solve computational problems due to the prevalent exploitation of the gradient-based RNN. However, the overlong convergent time and the too large residual error restrict the widespread applications of the RNN model in solving time-varying problems. Further, a special type of RNN named zeroing neural network (ZNN) is able to solve the time-varying Sylvester equation, which breaks the limitations mentioned above, but fails to handle complex time-varying problems owing to the sharp increment of the calculated amount in matrix inversion involved. To remedy the limitation, a modified gradient-based RNN (MGRNN) model is proposed to generate more accurate computational solutions with less convergent time and adaptive coefficients for solving the time-varying Sylvester equation, which replaces the matrix inversion problem with the matrix transposition problem. Besides, theoretical analyses and mathematical verifications are presented to validate the efficiency and superiority of the proposed MGRNN model compared with the traditional gradient-based recurrent neural network (GRNN) and ZNN models. Furthermore, simulation experiments are conducted to substantiate the properties of the newly proposed MGRNN model for solving the time-varying Sylvester equation.

1. Introduction

The linear system equation finds its various applications in many areas, such as image reconstruction [1] and machine learning [2], which would be solved by intelligent computing more or less. In paper [3,4], the researchers investigate the resilient control of the multi-agent systems under faults on sensors and actuators and study the heterogeneous multi-agent systems, where all agents may have different dynamics. Solving linear system equations is often carried out as the first step for dealing with many scientific problems, most of which are transformed as below: given that $A \in C^{m\times n}$ and $b \in C^m$, find the solution vector $x \in C^n$ to satisfy $Ax = b$ [5]. Moreover, some compelling researches about the control of linear systems have been achieved, such as the paper [6], which applies a combination of the off-policy learning and experience replay for output regulation tracking control of continuous-time linear systems with the completely unknown dynamics. The Sylvester equation, formulated as a linear system equation, is one of the basic instruments to research linear system problems. It is more extensive and more universal than Lyapunov equation which is one of the fundamental approaches to study the stability of linear systems. For example, from the automatic control and design perspective, one of the design procedures for filters is to calculate the related Sylvester equation, which devotes access to disturbance characteristics of the matrix invariant subspace. Also, it often appears in mathematical and control theories [7–11], and has an ingeniously extensive range of applications in a progressively increasing number of phases and situations, such as disturbance decoupling [12,13], image reconstruction [1], and semi-unsupervised learning [14]. However, with the progressively vigorous evolvement of the modern applications, the phases and situations are becoming more and more sophisticated with time-varying features in real-time. Thus, the time-varying Sylvester equation is becoming the research hotspot in the practical applications. Generally

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speaking, Sylvester equation have two subdivisions: the stationary one and the non-stationary one. For the former subdivision which works without time-varying parameters, a classic algorithm named Bartels–Stewart is utilized to obtain the numerical solution, which gets the Schur form converted from the matrix equality and acquires the solution by back-substitution with finite iterations and the time complexity being $O(n^3)$ [15]. For the other subdivision which has time-vary parameters, it is difficult for Bartels–Stewart algorithm to face abrupt changes during each sampling period not to mention the much more variable situations in practice, and thus Bartels–Stewart algorithm is of low-efficiency characteristics, which results from the large calculation burden [7].

As a result, it is worth constructing an appropriate approach to deal with the dynamic Sylvester equation with the aid of the vigorous neural networks. Under the condition that the initial situation is randomly generated, a recurrent neural network (RNN) model, often formulated as an ordinary differential equation (ODE) system, evolves along a preferable direction to decrease the corresponding residual error with time until the achievement of a predefined accuracy on estimated equilibrium points [16].

Considering the superiority of the RNN, a sort of models based on the RNN have been widely applied in fields of scientific calculation [17–19], signal processing [20], control theory and engineering [16,21–24], etc. Particularly, the gradient-based recurrent neural network (GRNN) model is exploited for solving matrix equations [25,26], of which a typical one could be defined as $\dot{X} = -\gamma A^{\mathrm{T}}(AX - I)$, where X is the unknown matrix to be obtained, I is the identity matrix and γ is used to scale the convergent speed, with the residual error being $AX - I$ [16]. From the above descriptions, we naturally conclude that the handling problem is converted into a zero-finding problem, which forces the residual error to converge toward zero as soon as possible along the negative direction of the gradient. However, lacking the velocity compensation of time-varying parameters, the RNN model is relatively inefficient to encounter dynamic zero-finding problems whose residual error can not globally converge to zero [7].

The Zeroing neural network model, which is termed ZNN model for short [27–30], is presented by Zhang et al., and has been prevalently exploited to handle the non-stationary Sylvester equation, which is an excellent substitution for traditional GRNN models. The highlight of the ZNN model is that the estimation error is able to converge to zero with infinite time compared with traditional GRNN models [10,31,32]. To remedy the drawback of convergent time of the ZNN model, the sign-bi-power activation is proposed by Li et al., to assist the ZNN model in converging to the theoretical solution of dynamic Sylvester equations with finite time [7]. Hitherto, a range of activation functions are respectively adapted to construct various models, revealing the superiority of reducing the convergent time of the ZNN model dramatically. To some extent, the models based on the ZNN achieve the superior performance when dealing with the non-stationary Sylvester equations with time-varying parameters. Nevertheless, some inherent limitations of the ZNN models severely restrain themselves from being applied in more comprehensive fields. The ZNN approach, involving the matrix inversion operation in the online computing and one of the limitations being the calculation burden of the matrix inversion, may encounter failures if the corresponding matrix is near singularity, however, the traditional GRNN models and the related modifications do not involve the matrix inversion [28].

To cover the shortages mentioned above, we propose a modified gradient-based recurrent neural network (MGRNN) model with adaptive coefficients, which can not only eliminate the matrix inversion but converge rapidly with the great precision when solving the time-varying Sylvester equation [33–35]. The crucial step that simplifies the calculation processes of the MGRNN model is to replace the matrix inversion problem with the matrix transposition problem.

The remainder of this paper is organized into four sections. In Section 2, the problem formulation and the existing researches including conversion procedures of Sylvester equations are presented and drawbacks of traditional GRNN models when facing the non-stationary Sylvester equation are analyzed from the mathematical perspective. The MGRNN model with adaptive coefficients and elimination of matrix inversion is proposed in Section 3, accompanying theoretical analyses and mathematical verifications for the efficiency and superiority of the MGRNN model. In Section 4, simulation examples and the comparisons between the MGRNN model and the traditional GRNN model are conducted to substantiate the efficiency and superiority of the MGRNN model. Finally, the conclusions are drawn in Section 5 with the corresponding remarks presented. Before ending this section, the main contributions of this paper are shown as follows:

1) The MGRNN model with adaptive coefficients and elimination of matrix inversion is proposed in this paper, providing a new manner besides the traditional GRNN model and ZNN model for solving the non-stationary Sylvester equation, which can generate precise solutions with the rapid convergence performance compared with the traditional methods.
2) The efficiency and superiority of the proposed MGRNN model are substantiated by theoretical analyses and mathematical verifications compared with the traditional GRNN model. Simulation examples and comparisons verify the correctness and feasibility of theorems proposed in this paper.
3) Differing from the traditional GRNN model and ZNN model, the matrix-inversion-free method proposed in this paper simplifies the calculation by transforming the inversion problem into the matrix transposition problem, so as to adapt to complex and changeable dynamic situations in the modern industry.

2. Problem formulation and the existing solutions

As portrayed in Section 1, massive efforts are committed to promoting the development of solving the non-stationary Sylvester equation for largely reducing the residual error and achieving rapid convergence. In this section, we discuss the performance of the traditional GRNN model when facing the time-varying Sylvester equation with dynamic parameters in real-time, and demonstrate its superiority and inferiority through theoretical analyses and mathematical verifications.

2.1. Preliminaries on Sylvester equation

Take the following time-varying Sylvester equation into consideration:

$$A(t)X(t) - X(t)B(t) + C(t) = 0, t \in [t_0, t_{\mathrm{f}}], \tag{1}$$

where $A(t) \in \mathbb{R}^{m\times m}$, $B(t) \in \mathbb{R}^{n\times n}$, $C(t) \in \mathbb{R}^{m\times n}$, t_0 and t_{f} are the initial and final time for the solving process, respectively; and the time-varying coefficient matrices are assumed to be known or estimated. Meanwhile, we assume that the non-stationary Sylverster Eq. (1) has a unique solution in any time with the prerequisite that $A(t)$ and $B(t)$ have no eigenvalue in common[36]. Rearranging $C(t)$ in (1) to the right side yields

$$A(t)X(t) - X(t)B(t) = -C(t), t \in [t_0, t_{\mathrm{f}}]. \tag{2}$$

Then, by performing vectorization on both sides of (2), it is easy to get

$$\mathrm{vec}(A(t)X(t) - X(t)B(t)) = -\mathrm{vec}(C(t)), \tag{3}$$

where $\mathrm{vec}(\cdot)$ represents the vectorization of a matrix. According to the two properties of vec-operator: $\mathrm{vec}(M + N) = \mathrm{vec}(M) +$

$\text{vec}(N)$ and $\text{vec}(MXN) = (N^{\mathrm{T}} \otimes M)\text{vec}(X)$, we carry out the following transformation of the left part of Eq. (3):

$$\begin{aligned}
&\text{vec}(A(t)X(t) - X(t)B(t)) \\
&= \text{vec}(A(t)X(t)) - \text{vec}(X(t)B(t)) \\
&= \text{vec}(A(t)X(t)I_n) - \text{vec}(I_m X(t)B(t)) \\
&= (I_n^{\mathrm{T}} \otimes A(t))\text{vec}(X(t)) - (B^{\mathrm{T}}(t) \otimes I_m)\text{vec}(X(t)) \\
&= (I_n \otimes A(t))\text{vec}(X(t)) - (B^{\mathrm{T}}(t) \otimes I_m)\text{vec}(X(t)) \\
&= (I_n \otimes A(t) - B^{\mathrm{T}}(t) \otimes I_m)\text{vec}(X(t)),
\end{aligned} \tag{4}$$

where $I_n \in \mathbb{R}^{nm \times nm}$ and $I_m \in \mathbb{R}^{m^2 \times m^2}$ denote the identity matrices. The mark $\otimes$ is the Kronecker product. The Kronecker product of an $m \times n$ matrix M and an $r \times q$ matrix N, is an $mr \times nq$ matrix. $M \otimes N$ is defined as

$$M \otimes N = \begin{bmatrix} M_{11}N & M_{12}N & \dots & M_{1n}N \\ M_{21}N & M_{22}N & \dots & M_{2n}N \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ M_{m1}N & M_{m2}N & \dots & M_{mn}N \end{bmatrix}. \tag{5}$$

Then, substituting Eq. (3) into Eq. (4) yields a vectorized form of the time-varying Sylverster equation as

$$(I_n \otimes A(t) - B^{\mathrm{T}}(t) \otimes I_m)\text{vec}(X(t)) = -\text{vec}(C(t)). \tag{6}$$

For the convenience of calculation in the subsequent sections, we rewrite the time-varying Sylvester equation as

$$P(t)\boldsymbol{x}(t) = \boldsymbol{b}(t), t \in [t_0, t_f], \tag{7}$$

where $P(t) = (I_n \otimes A(t) - B^{\mathrm{T}}(t) \otimes I_m) \in \mathbb{R}^{mn \times mn}$, $\boldsymbol{x}(t) = \text{vec}(X(t)) \in \mathbb{R}^{mn \times 1}$, $\boldsymbol{b}(t) = -\text{vec}(C(t)) \in \mathbb{R}^{mn \times 1}$.

2.2. Performance analyses on the GRNN model

As mentioned above, we can define an error function as $\epsilon(t) = P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)$ according to Eq. (7) and force it globally to converge to zero. Thus, the problem of Eq. (7) is turned into a zero-finding problem. As the classical and effective solution to zero-finding for solving the dynamic matrix problem, the GRNN model employs the Frobenius norm of the error matrix as the performance index with the corresponding neural network evolving along the negative gradient-descent direction designed to force the error norm to converge to zero with time in the time-invariant cases [37,38].

The GRNN model is outstanding at dealing with intricate situations or questions with extensive applications in many aspects of science and technology. Nonetheless, the traditional GRNN model cannot carry out high-precision real-time processing for dynamic problems with time-varying parameters. For the dynamic case, the error norm cannot converge to zero even after infinite time due to the lack of the velocity compensation for dynamic parameters [39]. A classic GRNN method can be formulated as

$$\dot{\boldsymbol{x}}(t) = -\gamma P^{\mathrm{T}}(t)(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)), \tag{8}$$

where the constant γ is used to scale the convergence rate.

Theorem 1. *The upper bound of the residual error generated by the GRNN model (8) for solving time-varying Sylvester Eq. (7) globally converges to an estimated error value instead of zero.*

Proof. The error between the calculated value $\boldsymbol{x}(t)$ and the theoretical value $\boldsymbol{x}^*(t)$ is defined as $\epsilon(t) = \boldsymbol{x}(t) - \boldsymbol{x}^*(t)$, and then it is obvious that

$$\dot{\epsilon}(t) = \dot{\boldsymbol{x}}(t) - \dot{\boldsymbol{x}}^*(t). \tag{9}$$

Define $V(t) = \|\epsilon(t)\|_2^2/2$ ($\|\cdot\|_2$ means 2-norm). In the perspective of zero-finding problems, our objective is to make $V(t)$ globally converge to zero. The time derivative of $V(t)$ is computed as

$$\dot{V}(t) = \frac{\partial V(t)}{\partial t} = \epsilon^{\mathrm{T}}(t)\dot{\epsilon}(t) = \epsilon^{\mathrm{T}}(t)(\dot{\boldsymbol{x}}(t) - \dot{\boldsymbol{x}}^*(t)). \tag{10}$$

Next, substituting the GRNN model (8) into (10) leads to

$$\dot{V}(t) = \epsilon^{\mathrm{T}}(t)\big(-\gamma P^{\mathrm{T}}(t)(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)) - \dot{\boldsymbol{x}}^*(t)\big). \tag{11}$$

Furthermore, $P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)$ could be rewritten as $P(t)\big(\boldsymbol{x}(t) - P^{-1}(t)\boldsymbol{b}(t)\big)$, where

$$P^{-1}(t)\boldsymbol{b}(t) = \boldsymbol{x}^*(\boldsymbol{t}). \tag{12}$$

Substituting (12) into $P(t)\big(\boldsymbol{x}(t) - P^{-1}(t)\boldsymbol{b}(t)\big)$ leads to

$$P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t) = P(t)\big(\boldsymbol{x}(t) - \boldsymbol{x}^*(t)\big) = P(t)\epsilon(t). \tag{13}$$

Thus, as Eq. (11) is unfolded,

$$\dot{V}(t) = -\gamma\epsilon^{\mathrm{T}}(t)P^{\mathrm{T}}(t)P(t)\epsilon(t) - \epsilon^{\mathrm{T}}(t)\dot{\boldsymbol{x}}^*(t). \tag{14}$$

Let δ be the mininal eigenvalue of $P^{\mathrm{T}}(t)P(t)$, we have

$$-\gamma\epsilon^{\mathrm{T}}(t)P^{\mathrm{T}}(t)P(t)\epsilon(t) \le -\gamma\delta\epsilon^{\mathrm{T}}(t)\epsilon(t) = -\gamma\delta\|\epsilon(t)\|_2^2. \tag{15}$$

Then, letting $\lambda \ge \|\dot{\boldsymbol{x}}^*(t)\|_2$ and combining with (14) and (15), we have

$$\begin{aligned}
\dot{V}(t) &\le -\gamma\delta\|\epsilon(t)\|_2^2 + \lambda\|\epsilon^{\mathrm{T}}(t)\|_2 \\
&\le -\|\epsilon(t)\|_2\big(\gamma\delta\|\epsilon(t)\|_2 - \lambda\big).
\end{aligned} \tag{16}$$

Hence, the convergence of the two sub-case of function (16) can be summarized as follows:

- For the case of $\gamma\delta\|\epsilon(t)\|_2 - \lambda \ge 0$, $\dot{V}(t) \le 0$, one can readily educe that $V(t)$ is of convergence conspicuously.
- For the case of $\gamma\delta\|\epsilon(t)\|_2 - \lambda < 0$, $\dot{V}(t) > 0$, $V(t)$ is diverging. Meanwhile, we have a formula as

$$\|\epsilon(t)\|_2 < \frac{\lambda}{\gamma\delta} \tag{17}$$

According to the above formula (17), it is evident that $V(t)$ is divergent in a circle with a radius of $\lambda/\gamma\delta$. In this sense, $\dot{V}(t)$ approaches to the verge of this circle.

In summary, according to the above discussions, $V(t)$ is globally convergent. The proof is fulfilled. □

Remark 1. From the above discussions, it is obvious that $V(t)$ synthesized by the classical GRNN model (8) globally converges to the surface of a ball even after infinite time. In addition, depending on the formula (17), we can draw a conclusion that along with the varying-rate of the instantaneous situation in real-time, the residual error generated by the GRNN model (8) increases due to the positive correlations between $\epsilon(t)$ and λ. Finally, to improve the convergent rate, the value of the indicator γ can be increased. Whereas, the value of constant γ cannot be enough large resulted from the current hardware facilities and related technical means. As a result, heightening the value of constant γ is not an effective method to fasten the convergence rate. Therefore, necessary modifications need to be applied to the GRNN model (8).

3. The MGRNN model and theoretical analyses

In this section, the MGRNN model is proposed to solve the time-varying Sylvester equation with a more precise solution and faster convergence speed with theoretical analyses and mathematical testifications. Meanwhile, the comparison between the MGRNN model and the traditional ZNN model on convergence calculation is conducted.

3.1. The MGRNN model

In the presence of dynamic parameters, the traditional GRNN model (8) cannot converge to zero in a short time, which restricts its practical exercises in the technology and science fields. However, the traditional GRNN model (8) can achieve great performance with advantages in many areas. Thus, it is worthwhile to refine the conventional GRNN model (8) into a modified form with adaptive coefficients as follows:

$$\dot{\boldsymbol{x}}(t) = -\varphi(t)\gamma P^{\mathrm{T}}(t)(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)), \tag{18}$$

where

$$\varphi(t) = \frac{c|\mathrm{vec}^{\mathrm{T}}\big(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)\big)\mathrm{vec}\big(\dot{P}(t)\boldsymbol{x}(t)\big)|}{\|P(t)\mathrm{vec}\big(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)\big)\|_2^2} \tag{19}$$

and $c > 1/\gamma$ denotes a constant. Besides, we construct a new function

$$\boldsymbol{\upsilon}(\boldsymbol{x}(t)) = \mathrm{vec}\big(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)\big). \tag{20}$$

Simultaneously, the MGRNN model (18) can be rewritten as

$$\dot{\boldsymbol{x}}(t) = -\varphi(t)\gamma\Big(\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial\boldsymbol{x}^{\mathrm{T}}(t)}\Big)^{\mathrm{T}}\boldsymbol{\upsilon}(\boldsymbol{x}(t)), \tag{21}$$

where

$$\varphi(t) = \frac{c|\boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial t}|}{\|\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial\boldsymbol{x}^{\mathrm{T}}(t)}\boldsymbol{\upsilon}(\boldsymbol{x}(t))\|_2^2}. \tag{22}$$

Now, the focus of solving time-varying Sylvester equation is converted into dealing with a zero-finding problem in vectorial form.

To explain the feasibility of the this MGRNN model (21) from the implementation and computational point of view, some extensions about the computation complexity of it could be conducted as follows: Since $\boldsymbol{\upsilon}(\boldsymbol{x}(t)) = \mathrm{vec}\big(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)\big)$, in Eqs. (21) and (22), we have $\boldsymbol{\upsilon}(\boldsymbol{x}(t)) \in \mathbb{R}^{mn\times 1}$, $\boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t)) \in \mathbb{R}^{1\times mn}$, $\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))/\partial t = \mathrm{vec}\big(P(t)\big) \in \mathbb{R}^{mn\times 1}$ and $\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))/\partial\boldsymbol{x}^{\mathrm{T}}(t) \in \mathbb{R}^{mn\times mn}$.

Before analyzing the computational complexity of the MGRNN model (21), we define a floating-point operation as one addition, subtraction, multiplication, or division of two floating-point numbers and recall the following facts [40].

- The multiplication of a scalar and a vector in size l_1 requires l_1 floating-point operations.
- For the multiplication of a matrix and a vector, one of size $l_1 \times l_2$ and the other of size l_2 require $l_1(2l_2 - 1)$ floating-point operations.
- The inversion of a square matrix in size $l_1 \times l_1$ requires l_1^3 floating-point operations.

Based on the above facts, at every sampling time instant, firstly, computing $\boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\big(\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))/\partial t\big)$ requires $2mn - 1$ floating-point operations, computing $\big(\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))/\partial\boldsymbol{x}^{\mathrm{T}}(t)\big)\boldsymbol{\upsilon}(\boldsymbol{x}(t))$ requires $mn(2mn - 1)$ floating-point operations and computing $\|\cdot\|_2^2$ requires $2mn - 1$ floating-point operations. Further, computing $c(|\cdot|)/\cdot$ requires 3 floating-point operations either. So computing $\varphi(t)$ requires $2mn - 1 + mn(2mn - 1) + 2mn - 1 + 3 = (2mn - 1)(mn + 2) + 3$ floating-point operations. Similarly, computing $\big(\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))/\partial\boldsymbol{x}^{\mathrm{T}}(t)\big)^{\mathrm{T}} = P(t)$ requires 0 floating-point operations; computing $\varphi(t)\gamma$ requires 1 floating-point operations; computing $\big(\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))/\partial\boldsymbol{x}^{\mathrm{T}}(t)\big)^{\mathrm{T}}\boldsymbol{\upsilon}(\boldsymbol{x}(t))$ requires $mn(2mn - 1)$ floating-point operations and computing $\varphi(t)\gamma\big(\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))/\partial\boldsymbol{x}^{\mathrm{T}}(t)\big)^{\mathrm{T}}\boldsymbol{\upsilon}(\boldsymbol{x}(t))$ requires $mn(2mn - 1) + 2$ floating-point operations. Thus, the computation complexity of the MGRNN model totally requires $(2mn - 1)(mn + 2) + 3 + mn(2mn - 1) + 2 + 1 = (2mn - 1)(2mn + 2) + 6$ floating-point operations at each time instant.

3.2. The traditional ZNN model

As manifested previously in the introduction section, according to Eq. (1), we define an error matrix as

$$E(t) = A(t)X(t) - X(t)B(t) + C(t), t \in [t_0, t_f]. \tag{23}$$

Similarly, from Eq. (7), we have the vectorization form of Eq. (23) as

$$e(t) = P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t), t \in [t_0, t_f], \tag{24}$$

where $P(t) = (I_n \otimes A(t) - B^{\mathrm{T}}(t) \otimes I_m) \in \mathbb{R}^{mn\times mn}$, $\boldsymbol{x}(t) = \mathrm{vec}(X(t)) \in \mathbb{R}^{mn\times 1}$ and $\boldsymbol{b}(t) = -\mathrm{vec}(C(t)) \in \mathbb{R}^{mn\times 1}$. Then, the traditional ZNN model is designed as

$$\dot{e}(t) = -\kappa\Psi(e(t)). \tag{25}$$

where $\kappa > 0$ denotes an index reflecting the convergence speed. Function $\Psi(\cdot)$ denotes the monotonically increasing odd activation function. Combining Eqs. (24) and (25) leads to

$$\dot{e}(t) = -\kappa\Psi\big(e(t)\big) = -\kappa\Psi\big(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)\big). \tag{26}$$

Depending on the properties of the ZNN model (25), the error vector (26) can converge to zero with infinite time and with a random initial condition $\boldsymbol{x}(0)$, in which $\boldsymbol{x}(t)$ is the estimated solution to the non-stationary Sylvester equation (1). Concerning a special case for the non-stationary Sylvester equation (1) of $B(t) \equiv 0$ and $C(t) \equiv -1$, the Sylvester equation $A(t)X(t) - X(t)B(t) + C(t) = 0$ converts into a form as $A(t)X(t) = I$, the theoretical solution of which is the time-varying inverse $A^{-1}(t)$ [27]. In addition, the corresponding matrix in the ZNN model (26) is $P(t)$. That is to say, calculating the inverse of a time-varying matrix $P^{-1}(t)$ is inevitable. On one hand, it is explicit to infer that it is unnecessary to calculate the inversion of time-varying matrix but just the matrix transposition by utilizing the MGRNN (21). Particularly, applying the Kronecker product makes the dimension surge sharply in short time, which numerously makes the computational amount of solving the time-varying matrix inversion costly. On the other hand, to solve the time-varying matrix transposition is much simpler and has no relationship with Kronecker product. Thus, the aim of eliminating the matrix inversion in the calculating procedures is achieved by the MGRNN (21). From the perspective of solving the non-stationary Sylvester equation, the traditional ZNN model (25) is not the sole solving manner. The MGRNN model can generate accurate computational solution rapidly with less computational amount for solving the time-varying Sylvester equation by replacing the matrix inversion problem with the matrix transposition.

3.3. Global convergence of the MGRNN model

In this section, the convergent efficiency of the MGRNN model (18) is demonstrated by the following presentation.

Theorem 2. *The residual error synthesized by the MGRNN model (18) for solving the time-varying Sylvester Eq. (1)* globally *converges to zero.*

Proof. Firstly, we define a Lyapunov function candidate as

$$U(t) = \boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\boldsymbol{\upsilon}(\boldsymbol{x}(t))/2. \tag{27}$$

For the case of $\boldsymbol{\upsilon}(\boldsymbol{x}(t)) \neq 0$, $U(t) > 0$ and $\boldsymbol{\upsilon}(\boldsymbol{x}(t)) = 0$, $U(t) = 0$, the conclusion that the above Lyapunov function candidate is positive definite can be drawn obviously. Thus, we have the time derivative of $U(t)$ as

$$\dot{U}(t) = \boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\left(\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial\boldsymbol{x}^{\mathrm{T}}(t)}\dot{\boldsymbol{x}}(t) + \frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial t}\right). \tag{28}$$

Substituting the $\dot{\boldsymbol{x}}(t)$ in the above Eq. (28) into Eq. (21) leads to

$$\dot{U}(t) = -\varphi(t)\gamma\boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial\boldsymbol{x}^{\mathrm{T}}(t)}\left(\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial\boldsymbol{x}^{\mathrm{T}}(t)}\right)^{\mathrm{T}}\boldsymbol{\upsilon}(\boldsymbol{x}(t)) + \boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial t}. \tag{29}$$

According to the property of norm as

$$A(t)A(t)^{\mathrm{T}} = \|A(t)\|_2^2 \tag{30}$$

we have

$$\begin{aligned}\dot{U}(t) &= -\varphi(t)\gamma\|\boldsymbol{\upsilon}(\boldsymbol{x}(t))\|_2^2\|\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial\boldsymbol{x}^{\mathrm{T}}(t)}\|_2^2 + \boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial t}\\ &= -\varphi(t)\gamma\|\boldsymbol{\upsilon}(\boldsymbol{x}(t))\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial\boldsymbol{x}^{\mathrm{T}}(t)}\|_2^2 + \boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial t}.\end{aligned} \tag{31}$$

Furthermore, replacing $\varphi(t)$ in Eq. (31) with Eq. (22), we have

$$\dot{U}(t) = -c\gamma|\boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial t}| + \boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial t}. \tag{32}$$

A verdict that $\dot{U}(t) < 0$ can be figured out categorically in terms of the case of $c > 1/\gamma$. The proof is completed. □

Remark 2. It is substantial to generalize that the residual error synthesized by the MGRNN model (18) globally converges to the theoretical solution on the basis of the Lyapunov function candidate in real-time. In other word, the MGRNN model (18) successfully solves the dynamic Sylvester problem with time-varying parameters.

3.4. The MGRNN model with activation functions for accelerating convergence

To accelerate the convergence speed of the proposed MGRNN model (18), different activation functions are exploited in this part. First of all, we define an activation function aided MGRNN model as

$$\dot{\boldsymbol{x}}(t) = -\varphi(t)\gamma P^{\mathrm{T}}(t)f(P(t)\boldsymbol{x}(t) - \boldsymbol{b}(t)), \tag{33}$$

where $f(\cdot)$ denotes the activation function including the linear, power-sigmoid and hyperbolic sine activation function. That is, $f_{\mathrm{li}}(\upsilon_i)$, $f_{\mathrm{ps}}(\upsilon_i)$ and $f_{\mathrm{hs}}(\upsilon_i)$ denote the linear, the power-sigmoid and the hyperbolic sine activation function, respectively, in the following discussion. Generally speaking, the following three activation functions are used to activate the MGRNN model (18) [41]: the linear (li) activation function:

$$f_{\mathrm{li}}(\upsilon_i) = \upsilon_i;$$

the power-sigmoid (ps) activation function :

$$f_{\mathrm{ps}}(\upsilon_i) = \begin{cases}\frac{1+\exp(-4)}{1-\exp(-4)}\frac{1-\exp(-4\upsilon_i)}{1+\exp(-4\upsilon_i)}, & \text{if } |\upsilon_i| < 1,\\ \upsilon_i^3, & \text{if } |\upsilon_i| \ge 1;\end{cases}$$

the hyperbolic sine (hs) activation function:

$$f_{\mathrm{hs}}(\upsilon_i) = \frac{\exp(3\upsilon_i)}{2} - \frac{\exp(-3\upsilon_i)}{2}.$$

In this section, we analyze the performance of convergence speed of the MGRNN model (18) activated by the linear and the nonlinear activation functions with the following theorem.

Theorem 3. *When activated by the nonlinear activation function, for example, the power-sigmoid or the hyperbolic sine activation function, the convergent efficiency of the MGRNN model (18) is more superior than the one activated by the linear activation function.*

Proof. We obtain a further conversion of the Lyapunov function candidate (27) as

$$U(t) = \boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\boldsymbol{\upsilon}(\boldsymbol{x}(t))/2 = \|\boldsymbol{\upsilon}(\boldsymbol{x}(t))\|_2^2/2 = \sum_{i=1}^{n}\upsilon_i^2(\boldsymbol{x}(t),t)/2. \tag{34}$$

Accordingly, we get the corresponding conversion of time derivative of $U(t)$ as

$$\begin{aligned}\dot{U}(t) &= \boldsymbol{\upsilon}^{\mathrm{T}}(\boldsymbol{x}(t))\left(\frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial\boldsymbol{x}^{\mathrm{T}}(t)}\dot{\boldsymbol{x}}(t) + \frac{\partial\boldsymbol{\upsilon}(\boldsymbol{x}(t))}{\partial t}\right)\\ &= \sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)\frac{\mathrm{d}\upsilon_i(\boldsymbol{x}(t),t)}{\mathrm{d}t}\\ &= -\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f(\upsilon_i(\boldsymbol{x}(t),t)),\end{aligned} \tag{35}$$

In this paper, we use these three activation functions to demonstrate the superior convergent efficiency of the MGRNN model (18). Firstly, to discuss the power-sigmoid activation function, we divide the analyses into two subsections for the value range of the $\upsilon_i(\boldsymbol{x}(t))$.

- For the case of $|\upsilon_i(t)| \ge 1$, $f_{\mathrm{ps}}(\upsilon_i)$ could be defined as the power function such as $f_{\mathrm{ps}}(\upsilon_i(t)) = \upsilon_i^3(t)$. Combining Eq. (34) with (35), we have

$$\begin{aligned}\dot{U}_{\mathrm{ps}}(t) &= -\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f_{\mathrm{ps}}(\upsilon_i(\boldsymbol{x}(t),t))\\ &= -\varphi(t)\sum_{i=1}^{n}\upsilon_i^4(\boldsymbol{x}(t),t)\\ &< -\varphi(t)\sum_{i=1}^{n}\upsilon_i^2(\boldsymbol{x}(t),t)\\ &= -\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f_{\mathrm{li}}(\upsilon_i(\boldsymbol{x}(t),t))\\ &= \dot{U}_{\mathrm{li}}(t),\end{aligned} \tag{36}$$

  where $\dot{U}_{\mathrm{li}}(t)$, $\dot{U}_{\mathrm{ps}}(t)$ and $\dot{U}_{\mathrm{hs}}(t)$ (in the posterior text) means that $\dot{U}(t)$ is activated by the linear, power-sigmoid and hyperbolic sine activation functions, respectively. From Eq. (36), we can draw a conclusion that the superior convergence is fulfilled for the MGRNN model (18) with adaptive coefficients by applying the power-sigmoid activation function compared with the one using the linear activation function.
- For the case of $|\upsilon_i(t)| < 1$, $f_{\mathrm{ps}}(\upsilon_i)$ could be defined as the bipolar-sigmoid function. Similarly, we have

$$\begin{aligned}\dot{U}_{\mathrm{ps}}(t) &= -\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f_{\mathrm{ps}}(\upsilon_i(\boldsymbol{x}(t),t))\\ &< -\varphi(t)\sum_{i=1}^{n}\upsilon_i^2(\boldsymbol{x}(t),t)\\ &= -\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f_{\mathrm{li}}(\upsilon_i(\boldsymbol{x}(t),t))\\ &= \dot{U}_{\mathrm{li}}(t).\end{aligned} \tag{37}$$

  From Eq. (37), we can also reach the same conclusion in the previous case. That is, the superior convergence is fulfilled by the MGRNN model (18) with the power-sigmoid activation function compared with the one activated by the linear activation function.

In addition, at the outset of the argument of the hyperbolic sine situation, we define the $\upsilon_i(\boldsymbol{x}(t),t)$ as $3\upsilon_i(\boldsymbol{x}(t),t)$ primarily. Then, exploiting Taylor expansion to $f_{\mathrm{hs}}(\upsilon_i(t),t)$ leads to

$$\begin{aligned} f_{\mathrm{hs}}(\upsilon_i(t),t) &= \left(\frac{\exp(3\upsilon_i)}{2} - \frac{\exp(-3\upsilon_i)}{2}\right) \\ &= \frac{1}{2}\left(\sum_{\kappa=1}^{+\infty}\frac{(3\upsilon_i)^\kappa}{\kappa!} - \sum_{\kappa=1}^{+\infty}\frac{(-3\upsilon_i)^\kappa}{\kappa!}\right) \\ &= \sum_{\kappa=1}^{+\infty}\frac{(3\upsilon_i)^{2\kappa-1}}{(2\kappa-1)!}\begin{cases} > 3\upsilon_i, \text{ if } \upsilon_i > 0 \\ < 3\upsilon_i, \text{ if } \upsilon_i < 0 \end{cases} \end{aligned} \tag{38}$$

- For the case of $\upsilon_i > 0$, integrating Eqs. (35) and (38), we have

$$\begin{aligned} \dot{U}_{\mathrm{hs}}(t) &= -\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f_{\mathrm{hs}}(\upsilon_i(\boldsymbol{x}(t),t)) \\ &< -\varphi(t)\sum_{i=1}^{n}3\upsilon_i^2(\boldsymbol{x}(t),t) \\ &= -3\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f_{\mathrm{li}}(\upsilon_i(\boldsymbol{x}(t),t)) \\ &= 3\dot{U}_{\mathrm{li}}(t) \end{aligned} \tag{39}$$

$$\begin{aligned} \dot{U}_{\mathrm{hs}}(t) &= -\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f_{\mathrm{hs}}(\upsilon_i(\boldsymbol{x}(t),t)) \\ &< -\varphi(t)\sum_{i=1}^{n}\upsilon_i(\boldsymbol{x}(t),t)f_{\mathrm{ps}}(\upsilon_i(\boldsymbol{x}(t),t)). \\ &= \dot{U}_{\mathrm{ps}}(t) \end{aligned} \tag{40}$$

- For the case of $\upsilon_i < 0$, similarly, we have the same conclusion as Eqs. (39) and (40).

In summary, integrating the above two subcases, it is plain to infer that Eqs. (39) and (40) are workable for any $|\upsilon_i(\boldsymbol{x}(t),t)| \neq 0$. That is, the achievement of the superior convergence is reached by exploiting the hyperbolic sine activation function for the MGRNN model (18) compared with the one using the linear activation function and the power-sigmoid activation function. Thus the proof is completed. □

Remark 3. On the basis of the Lyapunov function candidate, the smaller the value of time derivative $\dot{U}(t)$ is, the faster convergence speed can be. Because of the larger value of time derivative $\dot{U}_{\mathrm{li}}(t)$ using linear activation function compared with the one using nonlinear activation function (such as $\dot{U}_{\mathrm{ps}}(t)$ and $\dot{U}_{\mathrm{hs}}(t)$ in this paper), a superior convergent efficiency can be achieved by exploiting the nonlinear activation function. Furthermore, uniting the Theorem 2 projected previously, we could see that the MGRNN model (18) with adaptive coefficients using the activation function converges to the theoretical solution with a much more superior speed in the global scope. The convergence speed obtained by using the hyperbolic sine activation function is much superior than the ones obtained by utilizing the linear activation function or the power-sigmoid activation function.

4. Simulation examples and comparisons

So far, we have manifested the theoretical verification on the superiority and efficiency of the MGRNN model (18). In this section, simulation examples and comparisons are presented to substantiate the properties mentioned previously.

4.1. Example 1

In this example, we define an example of non-stationary Sylvester Eq. (1) with

$$\begin{cases} A(t) = \begin{bmatrix} \sin(t) & \cos(t) \\ -\cos(t) & \sin(t) \end{bmatrix}, \\ B(t) = \begin{bmatrix} 0.1\sin(t) & 0 \\ 0 & 0.2\cos(t) \end{bmatrix}, \\ C(t) = \begin{bmatrix} 0.1\sin^2(t) - 1 & -0.2\cos^2(t) \\ 0.1\sin(t)\cos(t) & 0.2\sin(t)\cos(t) - 1 \end{bmatrix}. \end{cases}$$

In this example, the $A(t)$ and the $B(t)$ have no eigenvalue in common. That is to say, the Sylvester equation has the unique solution assumption for any $C(t)$. The theoretical solution for this non-stationary Sylvester equation (1) is $X^*(t) = [\sin(t) - \cos(t); \cos(t)\sin(t)]$, which is given to verify the correctness of the solution produced by the MGRNN model (18). Meanwhile, we define the corresponding vectorized Eq. (7) with

$$\begin{cases} P(t) = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} \sin(t) & \cos(t) \\ -\cos(t) & \sin(t) \end{bmatrix}\right. \\ \left. - \begin{bmatrix} 0.1\sin(t) & 0 \\ 0 & 0.2\cos(t) \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right), \\ \boldsymbol{b}(t) = \begin{bmatrix} 0.1\sin^2(t) - 1 \\ 0.1\sin(t)\cos(t) \\ -0.2\cos^2(t) \\ 0.2\sin(t)\cos(t) - 1 \end{bmatrix}. \end{cases}$$

The corresponding simulation results are illustrated in Fig. 1. To validate the benefits and efficiency of the MGRNN model (18) more effectively, we set 8 random initial values with c=60 and $\gamma = 1$, which are depicted synthetically in Fig. 1(a). Specifically, the blue solid curves and red dash curves in Fig. 1(a) denote the generated trajectories of the computed solution $X(t)$ (i.e., $x(t)$ in vector form) synthesized by the MGRNN model (18) with adaptive coefficients and the theoretical solution $X^*(t)$ (i.e., $x^*(t)$ in vector form), respectively. From this figure, it is obvious that the blue trajectories of computed solution $X(t)$ generated by the MGRNN model (18) can converge to the red trajectories of theoretical solution $X^*(t)$.

To compare the convergent efficiency between MGRNN model (18) (blue solid curves) with adaptive coefficients and traditional GRNN model (8) (red dash curves) for solving the non-stationary Sylvester equation, the residual error $\|\epsilon(t)\|_{\mathrm{F}} = \|X(t) - X^*(t)\|_{\mathrm{F}}$ is utilized to show the convergence rate with the same parameters and random initial states in Fig. 1(b). Considering c=60 and γ=1 for all the models, we could have the following conclusions. Firstly, the residual error $\|\epsilon(t)\|_{\mathrm{F}} = \|X(t) - X^*(t)\|_{\mathrm{F}}$ generated by the traditional GRNN model (8) with the same 8 random initial values to MGRNN model (18) converges to around 1 in less than 3.5 s. However, the residual error $\|\epsilon(t)\|_{\mathrm{F}} = \|X(t) - X^*(t)\|_{\mathrm{F}}$ generated by the traditional GRNN model (8) converges to a constant value even after 10 s, which verifies Theorem 1. Secondly, the residual error $\|\epsilon(t)\|_{\mathrm{F}} = \|X(t) - X^*(t)\|_{\mathrm{F}}$ generated by MGRNN model (18) with 8 random initial values converges below 0.02 in less than 0.32 s, which verifies Theorem 2.

The note that the convergent efficiency of MGRNN model (18) with adaptive coefficients could be expedited by increasing the value of constant c, is demonstrated in Fig. 1(c) and (d). The trajectories denote $\|\epsilon(t)\|_{\mathrm{F}} = \|X(t) - X^*(t)\|_{\mathrm{F}}$ synthesized by

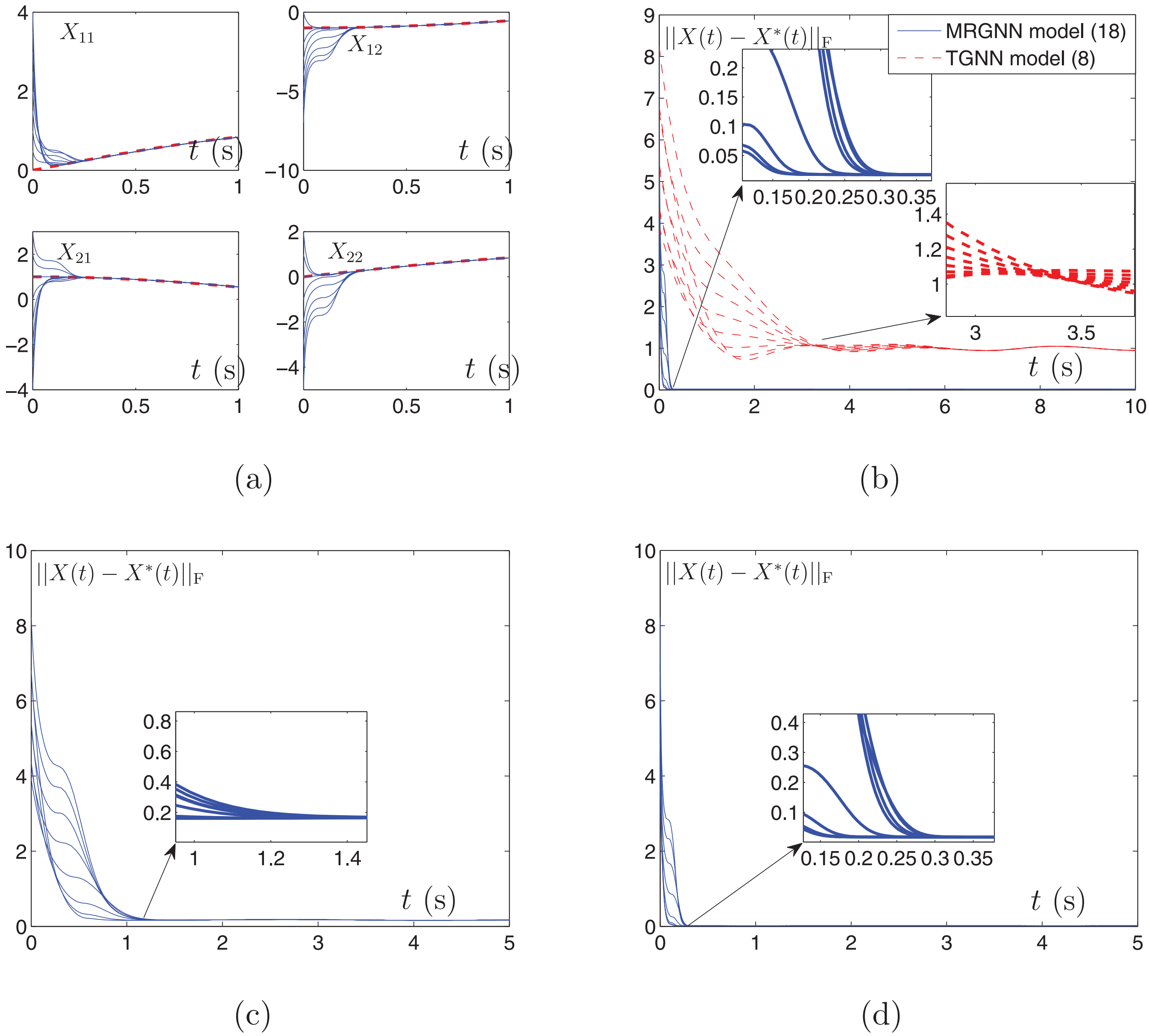


Fig. 1. Performance of MGRNN model (18) with $\gamma = 1$ and different value of c, where the 8 initial states are randomly generated, for computing the time-varying Sylvester Eq. (1). (a) The theoretical solution and computed solution of MGRNN model (18). (b) Performance of $||X(t) - X^*(t)||_F$ of MGRNN model (18) and that of traditional GRNN model (8). (c) Performance of $||X(t) - X^*(t)||_F$ of MGRNN model (18) with $\gamma = 1,\ c = 6$. (d) Performance of $||X(t) - X^*(t)||_F$ of MGRNN model (18) with $\gamma = 1,\ c = 60$.

the MGRNN model (18) with $c = 6$ and with $c = 60$, respectively, choosing $\gamma = 1$. As observed from Fig. 1(c), the 8 residual error trajectories denoting 8 random initial states exploited by MGRNN model (18) with $c = 6$ is below 0.2 after 1.3s approximately. In addition, the ones with $c = 60$ are much closer to zero than the former accompanying a much faster convergence speed (within about 0.3 s) in Fig. 1(d). Accordingly, one can readily deduce that different values of constant c in Eq. (19) can gain different convergent efficiencies.

For comparison, the convergent efficiency of the nonlinear function activated MGRNN model (18) with adaptive coefficients is shown in Fig. 2, in which the blue solid curves and red dash curves indicate that the MGRNN model (18) is activated by the power-sigmoid and the hyperbolic sine, respectively. From this figure, we could observe the differentiation of convergent efficiency with different values of constant c in Eq. (19) and the $\|\epsilon(t)\|_F = \|X(t) - X^*(t)\|_F$ of the nonlinear function activated MGRNN model (18) with $c = 6$ and $c = 60$ are shown in Fig. 2(a) and (b), respectively. It can be seen from the Figs. 1 and 2, the convergent efficiency of the MGRNN model (18) with adaptive coefficients activated by the nonlinear activation function is obviously better than that activated by the linear function. In addition, the one activated by the hyperbolic sine performs the best among the three activation functions as described in Theorem 3. Moreover, the residual error $\|\epsilon(t)\|_F = \|X(t) - X^*(t)\|_F$ utilizing the same nonlinear activation function approaches to zero within less time with $c = 60$ than that with $c = 6$, which is coincident with the remarks presented previously.

4.2. Example 2

To consider a more general case, we define an example of non-stationary Sylvester equation $A(t)X(t) - X(t)B(t) + C(t) = 0$,

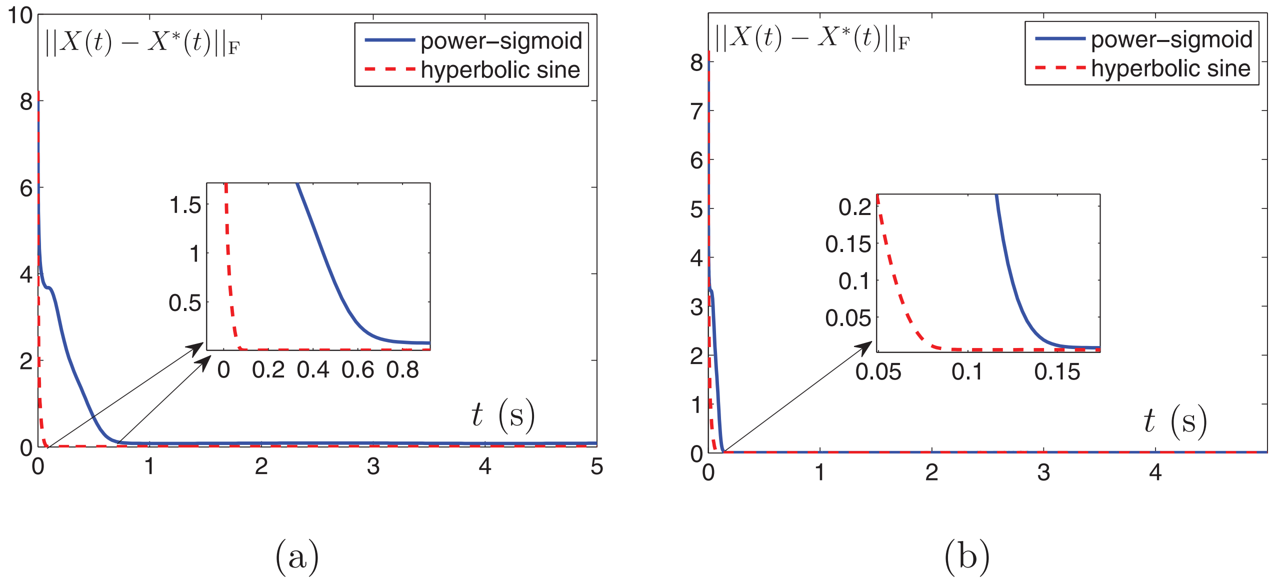


Fig. 2. Residual errors of nonlinear function activated MGRNN model (18) with adaptive coefficients, with $\gamma = 1$ and different values of c, for solving the time-varying Sylvester equation (1). (a) Performance of $||X(t) - X^*(t)||_\mathrm{F}$ of nonlinear function activated MGRNN model (18) with $\gamma = 1, c = 6$. (b) Performance of $||X(t) - X^*(t)||_\mathrm{F}$ of nonlinear function activated MGRNN model (18) with $\gamma = 1, c = 60$.

in which the time-varying matrices are not bounded.

$$\begin{cases} A(t) = \begin{bmatrix} \exp(t) & \exp(-t) \\ -\exp(-t) & \exp(t) \end{bmatrix}, \\ B(t) = \begin{bmatrix} 0.1\exp(t) & 0 \\ 0 & 0.2\exp(-t) \end{bmatrix}, \\ C(t) = \begin{bmatrix} -0.9\exp(2t) - \exp(-2t) & -0.2\exp(-2t) \\ 0.1 & 0.2 - \exp(-2t) - \exp(2t) \end{bmatrix}. \end{cases}$$

Meanwhile, we define the corresponding vectorized equation $P(t)x(t) = b(t)$ with

$$\begin{cases} P(t) = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} \exp(t) & \exp(-t) \\ -\exp(-t) & \exp(t) \end{bmatrix} - \begin{bmatrix} 0.1\exp(t) & 0 \\ 0 & 0.2\exp(-t) \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right), \\ \boldsymbol{b}(t) = \begin{bmatrix} -0.9\exp(2t) - \exp(-2t) \\ 0.1 \\ -0.2\exp(-2t) \\ 0.2 - \exp(-2t) - \exp(2t) \end{bmatrix}. \end{cases}$$

In this example, the $A(t)$ and the $B(t)$ have no eigenvalue in common. That is to say, the Sylvester equation has the unique solution assumption for any $C(t)$. The theoretical solution to this non-stationary Sylvester equation $A(t)X(t) - X(t)B(t) + C(t) = 0$ is $X^*(t) = [\exp(t) - \exp(-t); \exp(-t)\exp(t)]$, which is given to verify the correctness of the solution produced by the MGRNN model (18). The corresponding simulation results are illustrated in Fig. 3. Similarly, we set 8 random initial values with c=60 and $\gamma = 1$, which are depicted synthetically in Fig. 3(a). Specifically, the blue solid curves and red dash curves in Fig. 3(a) denote the generated trajectories of computed solution $X(t)$ (i.e., $x(t)$ in vector form) synthesized by the MGRNN model (18) with adaptive coefficients and the theoretical solution $X^*(t)$ (i.e., $x^*(t)$ in vector form), respectively. From this figure, we could draw the conclusion that the blue trajectories of computed solution $X(t)$ generated by the MGRNN model (18) are of convergence to the red trajectories of theoretical solution $X^*(t)$ rapidly, which is consistent with the simulation results of the bounded example carried out previously.

To manifest the convergent efficiency of the traditional GRNN model (red solid curves) for solving non-stationary Sylvester equation for this unbounded example, the residual error $\|\epsilon(t)\|_\mathrm{F} = \|X(t) - X^*(t)\|_\mathrm{F}$ is used to show the convergence rate with $\gamma = 1$ and 8 random initial states in Fig. 3(b). We could observe that the residual error $\|\epsilon(t)\|_\mathrm{F} = \|X(t) - X^*(t)\|_\mathrm{F}$ generated by the traditional GRNN model with random initial values converges to around 1.5 in less than 1.4 s, which verifies Theorem 1.

The note that the convergent efficiency of MGRNN model (18) with adaptive coefficients could be expedited by increasing the value of constant c is demonstrated in Fig. 3(c) and (d). The trajectories denote $\|\epsilon(t)\|_\mathrm{F} = \|X(t) - X^*(t)\|_\mathrm{F}$ of the MGRNN model (18) with $c = 6$ and with $c = 60$, respectively, choosing $\gamma = 1$ just like example 1. As observed from Fig. 3(c), the 8 residual error trajectories denoting 8 random initial states exploited by the MGRNN model (18) with $c = 6$ converge to zero after 0.8s approximately. In addition, the ones with $c = 60$ converge to zero much more rapidly and smoothly (within about 0.1 s) in Fig. 3(d). Accordingly, we could draw the conclusion that different values of constant c in Eq. (19) can gain different convergent efficiencies. This convergent performance of the MGRNN model (18) verifies Theorem 2, which could draw the same conclusion as the bounded example operated previously.

For comparison, the convergent efficiency of the nonlinear function activated MGRNN model (18) with adaptive coefficients is presented in Fig. 4, in which the blue solid curve and red dash curve indicate the MGRNN model (18) activated by the power-sigmoid and the hyperbolic sine, respectively. From this figure, we could observe the differentiation of convergent efficiency with different values of constant c in Eq. (19). And the residual errors $\|\epsilon(t)\|_\mathrm{F} = \|X(t) - X^*(t)\|_\mathrm{F}$ generated by the nonlinear function activated MGRNN model (18) with $c = 6$ and $c = 60$ are shown in Fig. 4(a) and (b), respectively. It can be seen from the Figs. 3 and 4, the convergent efficiency of the MGRNN model (18) with adaptive coefficients activated by the nonlinear activation function is better than that activated by the linear activation function. In addition, the one activated by the hyperbolic sine performs the

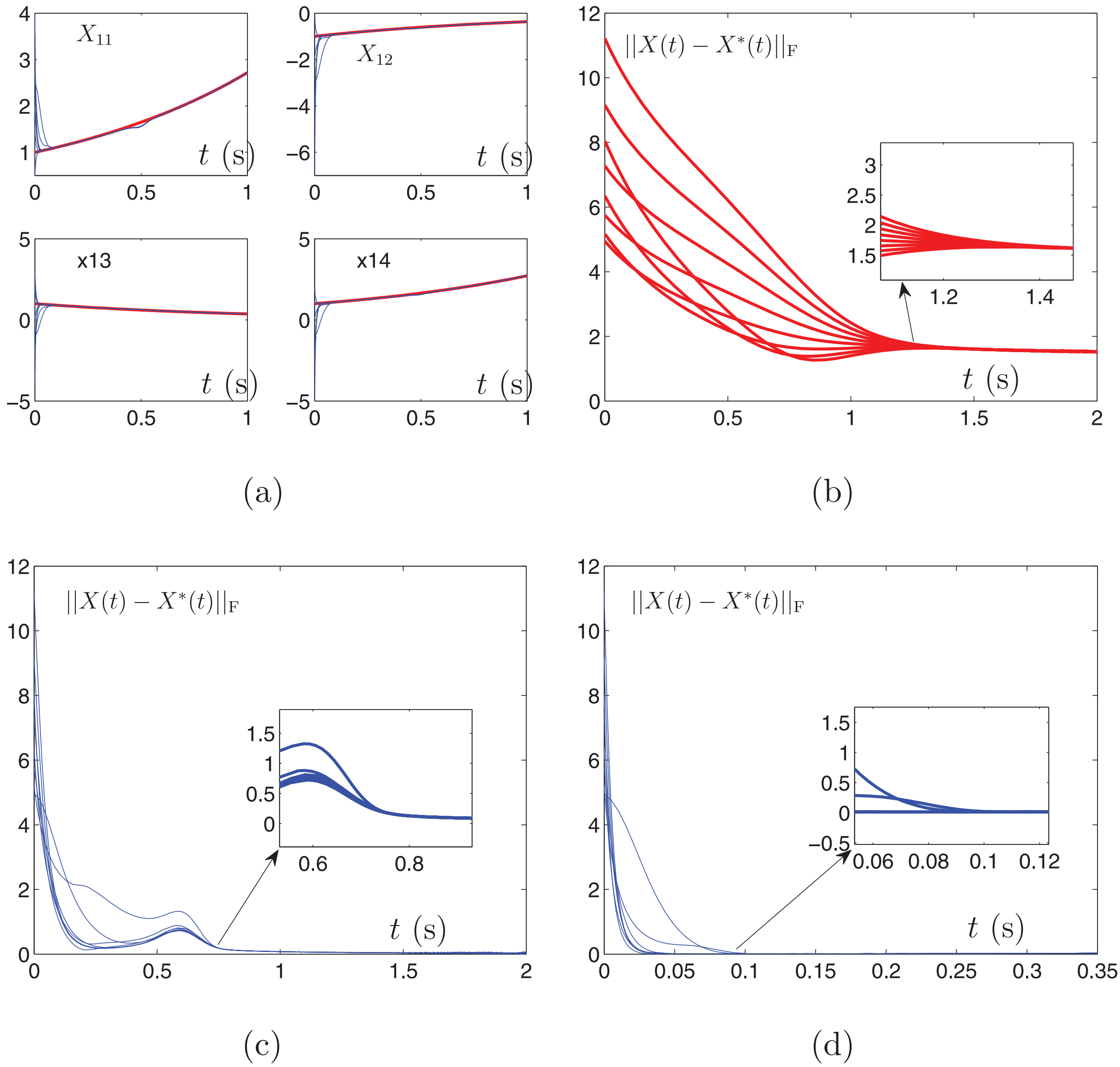


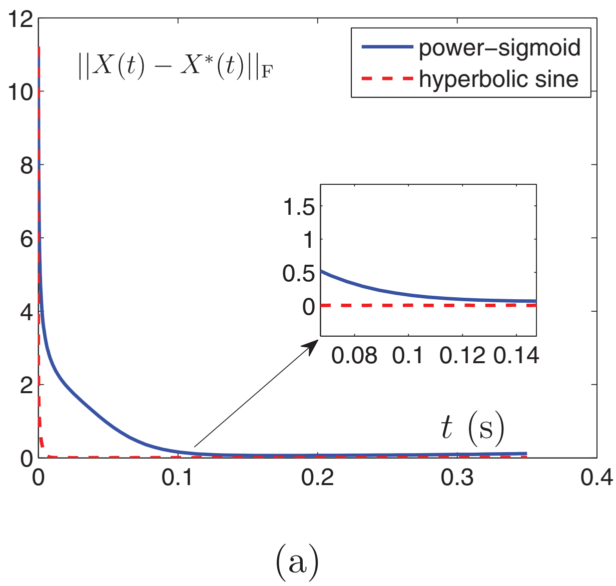
Fig. 3. Performance of MGRNN model (18) with $\gamma = 1$ and different value of c, where the 8 initial states are randomly generated, for computing the time-varying Sylvester equation. (a) The theoretical solution and computed solution of MGRNN model (18). (b) Performance of $\|X(t) - X^*(t)\|_F$ of traditional GRNN model. (c) Performance of $\|X(t) - X^*(t)\|_F$ of MGRNN model (18) with $\gamma = 1,\ c = 6$. (d) Performance of $\|X(t) - X^*(t)\|_F$ of MGRNN model (18) with $\gamma = 1,\ c = 60$.

Table 1
Comparisons Among the Traditional ZNN (26), the Traditional GRNN (8), and MGRNN (18) Models for Solving Time-varying Sylvester Equation.

	Theoretical Error	Globally Convergent	Free of Matrix Inversion	Designed from Control Perspective
Traditional GRNN (8)	Nonzero	Yes	Yes	No
Traditional ZNN (26)	Zero	Yes	No	Yes
MGRNN (18)	Zero	Yes	Yes	Yes

best among the three activation functions, which is corresponding to the Theorem 3. Moreover, the residual error $\|\epsilon(t)\|_F = \|X(t) - X^*(t)\|_F$ utilizing the same nonlinear activation function approaches to zero within less time with $c = 60$ than that with $c = 6$, which is also coincident with the remarks presented previously just as the bounded example.

The comparisons among the traditional ZNN (26), traditional GRNN (8), and MGRNN (18) models are summarized in Table 1. In the table, all these three models are of global convergence while only the theoretical error generated by the traditional GRNN model (8) cannot converge to zero. In addition, due to the lack of time derivative information [16], the traditional GRNN model (8) cannot deal with the dynamic problem with zero error. Moreover, computing the matrix inversion is inevitable for the traditional ZNN model (26) during the calculated procedures, which may lead to heavy calculation burden during the real-time computation, especially for



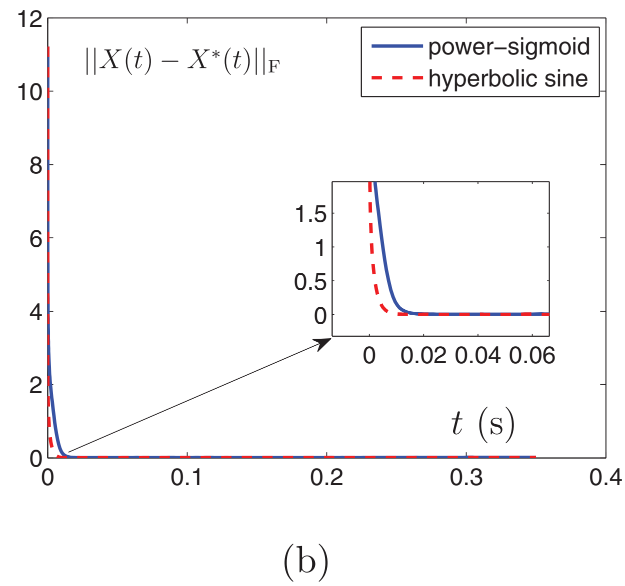


Fig. 4. Residual errors of nonlinear function activated MGRNN model (18) with adaptive coefficients, with $\gamma = 1$ and different values of c, for solving the time-varying Sylvester equation. (a) Performance of $||X(t) - X^*(t)||_F$ of nonlinear function activated MGRNN model (18) with $\gamma = 1, c = 6$. (b) Performance of $||X(t) - X^*(t)||_F$ of nonlinear function activated MGRNN model (18) with $\gamma = 1, c = 60$.

the near-singularity situation. The MGRNN model (18) with adaptive coefficients is able to handle the Sylvester equation with properties of inversion-free, error-free, and global convergence, which have been verified in previous sections by mathematic verifications, theoretical analyses, and simulation examples.

5. Conclusions

In this paper, we have revisited the traditional models, e.g., the ZNN model (25) and GRNN model (8), for solving the time-varying Sylvester equation and have pointed out the drawbacks of them through theoretical analyses. It has been concluded that the residual error synthesized by the GRNN model (8) converges to an estimated error even after infinite time. Besides, the calculation burden of the traditional ZNN model (25) has been pointed to be too large due to the involving of the time-varying matrix inversion. To remedy these drawbacks, the MGRNN model (18) has been proposed to solve the time-varying Sylvester equation with the adaptive coefficients and elimination of matrix inversion on the basis of mathematical verifications, which has the faster convergence rate and the more precise computed solution. At last, simulation examples and the comparisons have been provided to substantiate the convergent performance of the MGRNN model (18) by substituting the constant γ in the traditional GRNN model (8) with adaptive coefficients. Meanwhile, the matrix inversion-free manner of the proposed the MGRNN model (18) reduces the calculation burden of the time-varying matrix inversion in solving the time-varying Sylvester equation.

Declaration of Competing Interest

The authors declare that we have no relevant interest(s) to disclose.

Acknowledgments

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